

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations and computational treatments. Whether decomposing this heterogeneous task into multiple specialized agents improves end-to-end reliability over a single agent with equivalent capabilities remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system that assigns queries to four bounded specialists with scoped tool privileges and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid document retrieval with neural reranking for regulations. We evaluate CTU-Chat through three experimental scenarios on a 100-query held-out benchmark: (1) an architecture-level comparison against a functionally matched single-agent baseline and a routed generic-worker variant under identical model, knowledge, and tool capabilities; (2) architectural ablations that individually remove specialist policies, tool isolation, heterogeneous evidence allocation, deterministic calculation, and route repair; and (3) a semantic-neighbor tool-ambiguity stress test across registries of 11, 31, and 51 tools. In architecture comparison, CTU-Chat obtains an 8.0 pp higher End-to-End Success point estimate (81.0% vs. 73.0%; paired 95% CI [0.0, +16.0] pp; McNemar $p=0.096$) and 41.4% fewer input tokens at the cost of higher latency ($\sim 2\times$). Under registry expansion to 51 tools, the scoped specialist configuration achieves 88.0% E2E execution (vs. 78.7% for Top-10 selection), without displaying statistically divergent degradation slopes. These findings suggest that domain-scoped specialization and heterogeneous evidence allocation can support institutional counseling reliability under the evaluated conditions, with measurable trade-offs in coordination latency and token cost.

Keywords: Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) serves nearly 49,000 undergraduates in 2025–2026 [1]. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [10] and REBot for regulation advising [9]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [8,5,3], while agentic architectures support tool use and multi-agent coordination [20,18,16]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [14] further motivate deterministic calculation for financial queries.

Central hypothesis. When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with representation-specific evidence paths may improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges motivate three gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three contributions: **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; **C2**—an architecture-level comparative evaluation against a functionally matched single-agent baseline and a routed generic-worker variant; and **C3**—architectural ablations of five components and a semantic-neighbor tool-registry stress test across 11–51 tools.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, motivating coupling dense embeddings with exact lexical matching.

BM25 [15] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [7,8] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [2] merges BM25 and dense candidates without score calibration, while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG [6] demonstrates complexity-based retrieval selection. These approaches motivate dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [20] demonstrates interleaved reasoning and action, AutoGen [18] provides a multi-agent conversation framework, and Toolformer [16] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Importantly, controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [17]. CAAS supports admission counseling for IT programs [10], while REBot combines retrieval with category-guided GraphRAG for academic regulations [9]. CTU-CHAT departs from prior systems across three aspects: (1) *Task scope*: modeling 113 curricula as relational property graphs with deterministic arithmetic for tuition; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component-level ablations.

2.5 Research Gaps

Synthesizing the reviewed approaches identifies three gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

3 Proposed Model

This section presents CTU-CHAT’s architecture, algorithms, and evidence-acquisition mechanisms. The design follows three principles: *centralized agent coordination* (a contract-checked supervisor dispatches each query to one domain agent); *heterogeneous knowledge allocation* (each information structure maps to a corresponding evidence representation); and *representation-aware evidence routing* (graph-tool or intent-scoped retrieval path according to the assigned domain).

3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-CHAT specializes LangGraph through typed state (`AgentState` with validated rewrite, routes, evidence, and response), an intent–owner contract that checks structured supervisor output before dispatch, conditional routing (academic queries go directly to the graph agent; others through retrieval), acyclic orchestration (specialists return to END without peer edges), and asymmetric tool registries (each specialist sees only its scoped tools).

3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph contains over 1,500 nodes and 3,800 edges across 113 undergraduate curricula and fee schedules. Its 10 node labels cover curriculum entities (`Program`, `Course`, `Block`) and tuition entities (`TuitionFee`, `TuitionPolicy`, `ExemptionBasisRate`); 13 typed relationships encode containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names, comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

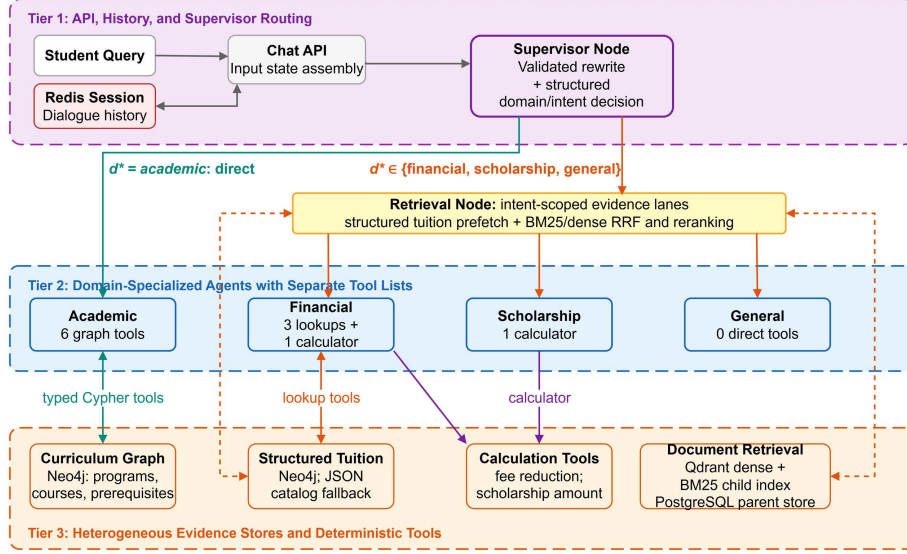


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

3.3 Routing and Deterministic Arithmetic

For a query q and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain d_{raw} and intent i_{raw} under a structured schema, falling back to `general`/other on parsing failure. The ownership contract repairs rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

Deterministic Tuition Arithmetic. For fee-reduction requests, the tool `tin_h_toan_hoc_phi` receives three parameters—per-credit tuition T , per-credit exemption basis B , and percentage p —and applies:

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (1)$$

The tool enforces $0 \leq p \leq 100$ and clamps P at zero. The contribution lies in composing validated rewriting, constrained routing, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

Rather than a uniform document index, CTU-CHAT preserves relational curricula in Neo4j, accesses cohort-dependent tuition through structured lookup, and retains narrative regulations in document retrieval. Academic queries enter the graph agent directly;

others enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback; the repaired intent then selects metadata-filtered document lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF: $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25, dense}\}} (60 + r_s(d))^{-1}$. The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

3.5 Cross-Domain Query Handling

A bounded specialist architecture raises the question of how queries spanning multiple domains are handled. CTU-CHAT adopts a *single-owner with multi-evidence prefetch* policy: the supervisor selects one primary owner based on dominant intent, but the retrieval layer collects evidence from all relevant sources before passing aggregated context to the owner. Specialists do not invoke other specialists; if the query requires a tool outside the owner’s scope, evidence is prefetched through retrieval rather than cross-specialist delegation. This design trades full cross-domain tool composability for bounded execution. The 20 cross-domain held-out queries directly test this mechanism.

4 Experimental Setup

4.1 Dataset and Annotation

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 56 single-source, 44 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

4.2 Baselines and Configurations

Fairness principle and execution contract. For Scenario 1, all architecture configurations share the same LLM (Gemini 2.5 Flash Lite, $T=0.0$), knowledge representations, retrieval stack (BM25 + dense + RRF + `bge-reranker-v2-m3`, $k=7$), and deterministic calculator access. The intended contrast is the *orchestration architecture*. Scenario 2 intentionally changes one component at a time. To reduce retrieval variance in Scenario 1,

all three architecture variants evaluate over the identical shared full-stack evidence retrieval output, yielding matched pre-generation Source Recall (78.6%) and Source AP (70.8%). The shared evidence context excludes outputs of tools whose invocation is evaluated; tool-derived results are available only after successful tool execution. Differences in final answers may be associated with the combined changes in tool privilege scoping, prompt specialization, and agent routing; Scenario 1 does not isolate their individual effects. Multi-agent variants use intent-routed scoped registries with identical execution bounds: maximum 3 tool calls, matching context limits, 2,048 output-token budget, and 30-second timeouts. Token accounting for CTU-CHAT aggregates all pipeline invocations (supervisor routing, contract repair, worker generation, and retries) for comparison with the single-agent baseline.

Scenario 1 — Architecture Comparison (RQ1): Three configurations are evaluated on the held-out set ($N=100$ queries, 300 runs under greedy decoding $T=0.0$): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

Scenario 2 — Ablation (RQ2): Five ablations each remove one component from Full CTU-Chat ($N=100$ queries per ablation, $T=0.0$): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

Scenario 3 — Tool Ambiguity (RQ3): Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding $50 \times 4 \times 3 \times 3 = 1,800$ decisions across counterbalanced blocks.

4.3 Evaluation Metrics

Metrics span four groups. *Final-answer quality:* Required-Fact Coverage (continuous proportion of reference facts verified in generated response), Answer Correctness, Answer Relevancy, and Faithfulness (Ragas [4]). *Evidence quality:* Source Recall, Source AP, Context Recall, Context Precision. *Execution reliability:* Tool Invocations per Query, Tool Selection Accuracy, Argument EM/Contract Match, Result Accuracy, and route repair rate. *Operational cost:* input/output tokens, LLM call counts, and wall-clock latency. In comparative evaluation, E2E Success Rate serves as the primary endpoint. Key secondary endpoints are Required-Fact Coverage, tool invocations, token cost, and latency. Source Recall and Source AP serve as controlled retrieval diagnostics, expected identical by design across Scenario 1 configurations.

E2E Success definition. For retrieval queries, E2E Success is evaluated under the criterion of required facts $\geq 50\%$, absence of unsupported claims, and retrieval of verified sources, complemented by continuous Fact Coverage. For tool queries: correct route,

tool, arguments, result, and answer; for the Unified Single Agent baseline, routing correctness is evaluated as N/A by construction as it does not employ a router. For adversarial queries: absence of unnecessary tool calls and appropriate clarification or refusal. For the arithmetic subset ($N=9$), numeric exact match against official bursar schedules is independently audited.

4.4 Statistical Protocol

Evaluations use greedy decoding ($T=0.0$) to minimize sampling variance. Paired differences across configurations are computed at the query level ($N=100$); 95% CIs are estimated using 10,000 bootstrap resamples. The primary confirmatory hypothesis is CTU-Chat versus Unified Single Agent on E2E Success. Generic-worker comparisons, continuous Fact Coverage, token cost, and latency are evaluated as secondary outcomes with reported effect sizes and 95% bootstrap CIs. Scenario 3 degradation is computed as $\Delta = \text{score}_{51} - \text{score}_{11}$ across counterbalanced tool-order blocks.

4.5 Infrastructure

Gemini 2.5 Flash Lite via Vertex AI ($T=0.0$). Neural reranking on a GTX 1650 (evaluation hardware). Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800-char, 100 overlap); Qdrant indexes child chunks (400-char, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. System uses bge-reranker-v2-m3, Lang-Graph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

5 Results

5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares the three configurations on the 100-query held-out benchmark. Because pre-generation evidence retrieval is held fixed across configurations, Source Recall (78.6%) and Source AP (70.8%) are identical across all three architectures. The observed differences in final answer quality and operational cost may be associated with the combined orchestration changes described in Section 4.

Table 1. Scenario 1: Architecture-level comparison ($N = 100$). Tools/Q = mean tool invocations (↓). Bold = best reliability / lowest cost.

Arch.	E2E	Fact	Src.R	Src.AP	Tools/Q	In Tok	Out Tok	Lat. (ms)
Single Agent	73.0	63.8	78.6	70.8	0.41	6296	1530	5159
Routed Gen.	71.0	59.6	78.6	70.8	0.31	3473	1415	8921
CTU-Chat	81.0	66.7	78.6	70.8	0.28	3690	2092	10652

Pairwise bootstrap analysis (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves an 8.0 pp higher E2E point estimate (81.0% vs. 73.0%, paired 95% bootstrap

CI [0.0, +16.0] pp; exact McNemar $p=0.096$), alongside higher Fact Coverage ($\Delta= +2.8$ pp, 95% CI [−2.1, +7.9] pp) and 41.4% fewer input tokens alongside 26.1% fewer total tokens ($\Delta= -2,044$ total tokens, CI [−2,710, −1,378]; Single: 7,826 vs. CTU: 5,782), at the cost of +5.49 s median end-to-end latency. (2) CTU-Chat vs. Routed Generic: specialist policies raise Fact Coverage by +7.05 pp (CI [+1.0, +13.2] pp, $p<0.05$). (3) Routed Generic vs. Single Agent: comparable token savings ($\Delta= -2,824$ input tokens) but a −4.2 pp Fact Coverage point estimate (CI [−9.8, +1.3] pp); the interval includes zero.

On the 20 cross-domain queries, CTU-Chat achieves an 85.0% E2E point estimate compared with 75.0% (Single) and 70.0% (Generic). A sensitivity check across fact-coverage thresholds shows that CTU-Chat has a higher point estimate at $\theta=0.50$ (81.0% vs. 73.0%; +8.0 pp) and $\theta=0.60$ (63.0% vs. 59.0%; +4.0 pp), while Single Agent has a slightly higher point estimate at $\theta=0.75$ (41.0% vs. 39.0%) and $\theta=1.00$ (33.0% vs. 32.0%).

5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ($N=100$). Deltas are relative to the full system.

Table 2. Scenario 2: Ablation results ($N = 100$). Negative Δ = degradation.

Ablated	Δ E2E	Δ Fact	Δ Src.R	Δ Tok	Δ Lat
No Spec. Policy	−10.0	−7.05	0.0	−217	−1731
Full Tool Vis.	−3.0	−0.02	0.0	+1270	+296
Uniform Text-RAG	−10.0	−2.67	−12.3	−300	−1910
No Route Rep.	−4.0	−2.05	0.0	−219	−1610
No Determ. Calc. [†]	−6.0	−0.97	0.0	−782	−4540

[†]Full benchmark with calculators disabled. Δ Tok/Lat in tokens/ms per query.

The two largest observed degradations occur in *No Specialist Policy* (Δ E2E = −10.0 pp, 95% CI [−19.0, −2.0] pp; Δ Fact = −7.05 pp, CI [−13.13, −1.22] pp) and *Uniform Text-RAG Baseline* (Δ E2E = −10.0 pp, CI [−16.0, −5.0] pp; Δ Src.R = −12.3 pp, CI [−16.5, −8.3] pp). Removing tool isolation causes a minor shift (Δ E2E = −3.0 pp, CI [−9.0, +3.0] pp, crossing zero) while expanding input tokens by +1,270. Disabling route repair yields Δ E2E = −4.0 pp (CI [−10.0, 0.0] pp), disproportionately harming cross-domain queries (−15.0 pp). Disabling deterministic calculation produces Δ E2E = −6.0 pp (CI [−12.0, −1.0] pp) across the benchmark; on the arithmetic tuition subset ($N=9$), exact-match answers drop from 8/9 to 3/9 (−55.6 pp exact match pass) due to unconstrained arithmetic hallucination.

5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 4 reports the semantic-neighbor experiment across 11–51 tools ($N=50$, 3 counter-balanced tool-order blocks, 1,800 primary decisions).

Table 3. Scenario 2: E2E Success by query family across ablations.

Config.	Graph ($n=19$)	Struct-fin ($n=28$)	Narr-reg ($n=33$)	Cross ($n=20$)
Full CTU-Chat	73.7	85.7	78.8	85.0
No Spec. Policy	63.2	82.1	66.7	70.0
Full Tool Vis.	78.9	71.4	78.8	85.0
Uniform Text-RAG	73.7	67.9	75.8	65.0
No Route Rep.	78.9	78.6	75.8	70.0

Table 4. Scenario 3: Semantic-neighbor tool ambiguity ($N = 50$, 3 counterbalanced tool-order blocks).

Reg.	Config.	Vis.	Gold-R	Sel.	E2E	Tok	Lat
11	Full-reg single	11	—	94.0	86.0	1663	1098
	Top-10 single	10	100.0	93.3	83.3	1541	1148
	Routed gen.	5.1	98.0	90.7	80.0	1204	1889
	Routed spec.	5.1	98.0	93.3	85.3	1441	1983
31	Full-reg single	31	—	91.3	84.7	2706	1157
	Top-10 single	10	98.0	90.0	81.3	1149	1033
	Routed gen.	9.8	98.0	88.0	79.3	1396	1958
	Routed spec.	9.8	98.0	93.3	88.7	1830	2015
51	Full-reg single	51	—	92.7	83.3	3722	1289
	Top-10 single	10	96.0	86.7	78.7	1036	997
	Routed gen.	9.8	98.0	91.3	82.7	1340	1907
	Routed spec.	9.8	98.0	92.7	88.0	1770	2021

As the registry scales from 11 to 51 tools, the Full-Registry Single Agent changes from 86.0% to 83.3% E2E while input tokens increase by 124%. The Global Top-10 changes from 83.3% to 78.7% E2E, alongside a Gold Recall change from 100.0% to 96.0%. The Routed Specialist records 88.0% E2E at 51 tools with 9.8 visible tools. At the 51-tool endpoint, Routed Specialist has a +9.3 pp E2E difference over Top-10 Single Agent (95% CI [+2.0, +17.3] pp) and a +4.7 pp point-estimate difference over Full-Registry Single Agent (95% CI [-1.3, +11.3] pp) with 52.4% fewer input tokens (1,770 vs. 3,722). The degradation interaction (Specialist vs. Top-10 high-minus-low difference) is +7.3 pp (95% CI [0.0, +15.3] pp).

5.4 Supporting Retrieval Validation

Table 5 validates the retrieval subsystem independently.

Table 5. Supporting evaluation: Progressive retrieval component stacking ($N = 100$).

Config.	Hit@1	Hit@3	P@5	R@5	MRR@10	Lat. (ms)
BM25 Lexical	0.810	0.920	0.234	0.937	0.874	14
Dense Semantic	0.630	0.840	0.202	0.853	0.736	80
Hybrid RRF	0.770	0.940	0.222	0.935	0.851	80
Hybrid + Reranker	0.830	0.940	0.230	0.955	0.896	1,145
Full Retrieval	0.850	0.960	0.234	0.975	0.913	1,153

BM25 records 0.810 Hit@1 and 0.874 MRR@10, compared with 0.630 Hit@1 and 0.736 MRR@10 for dense semantic search. Hybrid RRF records 0.940 Hit@3 and 0.770 Hit@1. Adding the cross-encoder reranker (`bge-reranker-v2-m3`) raises Hit@1 to 0.830 and MRR@10 to 0.896. The full retrieval configuration records 0.850 Hit@1 and 0.975 Recall@5. Because this final configuration changes multiple evidence components, the difference cannot be attributed to one component alone.

5.5 Failure Analysis

Table 6 categorizes error modes across the 46 failure instances (< 50% Fact Coverage) in Scenario 1 (27 for Single Agent, 19 for CTU-Chat).

CTU-Chat has 2 recorded tool errors versus 7 for Single Agent and 2 unnecessary calls versus 5. Gold-source absence and reranker-demotion counts are identical (2 and 4 respectively), while cross-domain partial omissions number 3 versus 5.

6 Discussion

6.1 Answering the Research Questions

RQ1 (Architecture-Level Effectiveness): The architecture-level comparison indicates that supervisor-routed multi-agent decomposition obtains an 8.0 pp higher E2E point

Table 6. Failure taxonomy across architecture configurations ($N = 100$). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

Category	Failure Type	Single CTU	
Routing	Misroute corrected by repair	N/A	4
	Unrecoverable misroute	N/A	2
Retrieval	Gold source absent	2	2
	Cross-domain partial omission	5	3
	Reranker demoted gold below $k=5$	4	4
Tool	Invalid/default arguments	4	1
	Wrong-family hallucination	3	1
	Unnecessary invocation	5	2
Generation	Tool result ignored in text	2	1
	Partial fact omission	11	9
Total	Fact Coverage < 50%	27	19

estimate than the Unified Single Agent (81.0% vs. 73.0%), although the paired bootstrap interval touches zero ([0.0, +16.0] pp; McNemar $p=0.096$). CTU-Chat also has a +2.8 pp continuous Fact Coverage point estimate (95% CI [-2.1, +7.9] pp) and 41.4% fewer input tokens alongside 26.1% fewer total tokens (3,690 vs. 6,296 input; 5,782 vs. 7,826 total), with higher mean latency (10,652 vs. 5,159 ms). The Routed Generic configuration has lower Fact Coverage than CTU-Chat (59.6% vs. 66.7%; $\Delta = -7.05$ pp, $p < 0.05$), suggesting that specialist policies may contribute to the observed performance under this setup.

RQ2 (Architectural Mechanism): The component ablations show distinct changes: uniform text retrieval (S2-A3) has the largest Source Recall drop (-12.3 pp, including -17.8 pp on structured finance); removing specialist prompts (S2-A1) has the largest Fact Coverage decline (-7.05 pp); full tool visibility (S2-A2) is associated with +1,270 input tokens and little coverage change; omitting route repair (S2-A5) corresponds to -15.0 pp on cross-domain queries; and disabling deterministic calculation (S2-A4) corresponds to a 55.6 pp lower exact-match rate on arithmetic queries. These patterns are consistent with different roles for the components in this benchmark.

RQ3 (Tool Ambiguity): At 51 tools, Routed Specialist records 88.0% E2E with 9.8 visible tools, whereas Full-Registry Single Agent records 83.3% with 124% more input tokens than at 11 tools, and Global Top-10 records 78.7%. The 51-tool endpoint favors the specialist over Top-10 in this stress test; the reported interaction interval does not establish different degradation slopes.

6.2 Relation to Prior Work

While ReAct [20] and Toolformer [16] explore tool use in single-agent settings, the tested full-registry configuration uses more input tokens at 51 than at 11 tools. Unlike general multi-agent frameworks (e.g., AutoGen [18]) that permit peer interaction, CTU-Chat

uses hierarchical, contract-validated routing with single-owner prefetch. The S2-A3 ablation shows lower source recall with uniform text retrieval in this benchmark, consistent with the motivation for structured evidence access in GraphRAG [3]. Whereas Adaptive-RAG [6] routes by semantic complexity, CTU-Chat routes by institutional representation requirements.

6.3 Threats to Validity

Internal: Single LLM (Gemini 2.5 Flash Lite); while greedy decoding minimizes sampling variance, API serving variability and prompt sensitivity remain potential confounds. **External:** Single-institution benchmark (100 queries); findings may not directly generalize to other institutional statutes. Semantic-neighbor tools are controlled evaluation constructs. **Construct:** Automated metrics rely partly on Ragas; programmatic required-fact matching reduces generator–judge bias. **Statistical:** Bootstrap CIs are query-level ($N=100$); arithmetic ablation ($N=9$) has small sample size, though 8/9 vs. 3/9 exact-match differences are substantial. **Operational:** Wall-clock latency reflects API serving variability and evaluation hardware.

7 Conclusion and Future Work

This study developed and evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for heterogeneous university counseling, through architecture-level comparison, mechanistic ablation, and tool-ambiguity stress testing.

Three principal findings address the research questions. For RQ1, the architecture-level comparison yields an 8.0 pp higher End-to-End Success point estimate for CTU-Chat than for a functionally matched single-agent baseline (81.0% vs. 73.0%), although the paired CI touches zero and McNemar $p=0.096$; CTU-Chat uses 41.4% fewer input tokens with higher latency. For RQ2, removing specialist policies corresponds to -10.0 pp E2E, while replacing heterogeneous evidence allocation corresponds to -12.3 pp Source Recall in this benchmark. For RQ3, the Routed Specialist records 88.0% E2E at 51 tools, compared with 83.3% for the Full-Registry Single Agent and 78.7% for Global Top-10.

These findings are consistent with the interpretation that supervisor-routed bounded specialization—comprising domain-specific policies, scoped tool access, representation-specific evidence paths, and deterministic computation—can support heterogeneous institutional counseling under the evaluated conditions, with measurable trade-offs in coordination latency and token cost.

Future work should (1) reduce routing latency through parallel candidate dispatch and prompt compression; (2) extend evaluation to multi-institutional and bilingual benchmarks with naturally evolving tool registries; (3) conduct human evaluation with multiple raters to complement automated metrics; and (4) perform field trials assessing student trust and advisory efficacy in live counseling settings.

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